

HSC Extension 2 Mathematics

201

Further Work with Vectors



The points A, B and C are collinear where

$$\overrightarrow{OA} = \hat{i} + \hat{j}, \quad \overrightarrow{OB} = 2\hat{i} - \hat{j} + \hat{k}, \quad \overrightarrow{OC} = 3\hat{i} + a\hat{j} + b\hat{k}.$$

What are the values of a and b ?

- A) $a = 3, b = 2$ B) $a = -3, b = 2$
 C) $a = 3, b = -2$ D) $a = -3, b = -2$



HSC Extension 2 Mathematics

202

Further Work with Vectors



If $\vec{a} = 3\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} - \hat{k}$, find the unit vector(s) perpendicular to both $\vec{a}$ and $\vec{b}$

2



HSC Extension 2 Mathematics

203

Further Work with Vectors



The points P, Q and R have position vectors $\hat{i} + 2\hat{j} - \hat{k}$, $\hat{i} + 2\hat{j}$ and $2\hat{i} - 3\hat{j} - 2\hat{k}$ respectively.

- (i) Find the vectors $\overrightarrow{QP}$ and $\overrightarrow{QR}$ 1
 (ii) Find the magnitude of $\angle PQR$ correct to the nearest degree 2



HSC Extension 2 Mathematics

204

Further Work with Vectors



(i) Find the vector equation of line vector $\vec{r}$ given it passes through $(1, 3, -2)$ and $(2, -1, 2)$ 1

(ii) Determine if $\vec{r}$ passes through $(4, -9, 10)$ 1



Further Work with Vectors Solution

Let the unit vector be $\underline{v} = a\underline{i} + b\underline{j} + c\underline{k}$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = 0 \quad \text{and} \quad \begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} = 0 \quad \text{gives}$$

$$\begin{aligned} 3a + 2b + 2c &= 0 \quad (1) \\ a - b - c &= 0 \quad (2) \\ (1) + 2 \times (2) & \rightarrow 5a = 0 \\ \therefore a &= 0 \quad \therefore b = -c \end{aligned}$$

Unit vector means $a^2 + b^2 + c^2 = 1$
 $\therefore 2b^2 = 1$
 $b = \pm \frac{1}{\sqrt{2}} \quad c = \mp \frac{1}{\sqrt{2}}$
 $\therefore \underline{v} = \pm \left(\frac{1}{\sqrt{2}} \underline{j} - \frac{1}{\sqrt{2}} \underline{k} \right) \quad (3)$



Further Work with Vectors Solution

$$\frac{2}{1} = \frac{a-1}{-2} = \frac{b}{1}$$

$$\therefore a = -3, \quad b = 2$$

B



Further Work with Vectors Solution

$$(i) \underline{r} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2-1 \\ -1-3 \\ 2-2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix} \quad (1)$$

since $\underline{r}$ is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix} \Rightarrow \begin{aligned} 4 &= 1 + \lambda \Rightarrow \lambda = 3 \\ -9 &= 3 - 4\lambda \Rightarrow \lambda = 3 \quad (1) \\ 10 &= -2 + 0\lambda \Rightarrow \lambda = 3 \end{aligned}$$

$\therefore$ Line can pass through $(4, -9, 10)$



Further Work with Vectors Solution

$$(i) \underline{QP} = \underline{i} + 2\underline{j} - \underline{k} - (\underline{i} + 2\underline{j}) \quad \underline{QR} = 2\underline{i} - 3\underline{j} - 2\underline{k} - (\underline{i} + 2\underline{j})$$

$$= -\underline{k} \quad = \underline{i} - 5\underline{j} - 2\underline{k}$$

$$(1) \text{ for both } = \underline{i} - 5\underline{j} - 2\underline{k}$$

$$(ii) \underline{QP} \cdot \underline{QR} = |\underline{QP}| |\underline{QR}| \cos \theta$$

$$2 = 1 \times \sqrt{1^2 + 5^2 + 2^2} \cos \theta \quad (1)$$

$$\cos \theta = \frac{2}{\sqrt{30}}$$

$$\theta = 69^\circ \quad (2)$$



HSC Extension 2 Mathematics

205

Further Work with Vectors



If $P = \underline{i} + \underline{j} + \underline{k}$ and $R = 9\underline{i} + 3\underline{j} + 8\underline{k}$ find the point Q on $\overline{PR}$ such that $PQ : QR = 2 : 3$ 3



HSC Extension 2 Mathematics

206

Further Work with Vectors



Find a point of intersection of the line with parametric equation $r = \underline{i} + 3\underline{j} - 4\underline{k} + t(\underline{i} + 2\underline{j} + 2\underline{k})$ and the sphere with equation $(x - 1)^2 + (y - 3)^2 + (z + 4)^2 = 81$ 3



HSC Extension 2 Mathematics

207

Further Work with Vectors



The point $P(x, y)$ representing the complex number z moves in the Argand diagram so that $|z - 6| = |z + 2i|$.

- (i) Show that the path P traces out, has equation $3x + y - 8 = 0$ 2
- (ii) Find the minimum value of $|z|$ as P moves on this path 2



HSC Extension 2 Mathematics

208

Further Work with Vectors



Given $\overline{AP} = \begin{bmatrix} -1 \\ 6 \\ -1 \end{bmatrix}$ and $\overline{AB} = \begin{bmatrix} -2 \\ 5 \\ -7 \end{bmatrix}$

- (i) Find $\text{proj}_{\underline{b}} \underline{p}$ where $\overline{AP} = \underline{p}$ and $\overline{AB} = \underline{b}$ 2
- (ii) Find the perpendicular distance d from P to the line AB . 2



$$\begin{aligned}
 x &= 1+t \\
 y &= 3+2t \\
 z &= -4+2t \quad (1)
 \end{aligned}$$

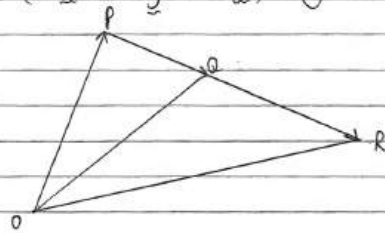
Sub. these into sphere

$$\begin{aligned}
 (1+t-1)^2 + (3+2t-3)^2 \\
 + (-4+2t+4)^2 &= 81 \\
 t^2 + (2t)^2 + (2t)^2 &= 81 \\
 9t^2 &= 81 \\
 t^2 &= 9 \\
 t &= \pm 3 \quad (2)
 \end{aligned}$$

sub. into (1) (2) (3)
gives $(4, 9, 2)$ or
 $(-2, -3, -10)$
(one of) (3)



$$\begin{aligned}
 \vec{PR} &= (9\mathbf{i} + 3\mathbf{j} + 8\mathbf{k}) - (\mathbf{i} + \mathbf{j} + \mathbf{k}) \\
 &= 8\mathbf{i} + 2\mathbf{j} + 7\mathbf{k} \\
 \vec{PQ} &= \frac{2}{5}(8\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}) \quad (1)
 \end{aligned}$$



$$\begin{aligned}
 \text{Now } \vec{OP} + \vec{PQ} &= \vec{OQ} \\
 \therefore \vec{OQ} &= \mathbf{i} + \mathbf{j} + \mathbf{k} + \frac{2}{5}(8\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}) \quad (2) \\
 \vec{OQ} &= \frac{21}{5}\mathbf{i} + \frac{7}{5}\mathbf{j} + \frac{17}{5}\mathbf{k} \\
 \therefore Q &= \frac{21}{5}\mathbf{i} + \frac{7}{5}\mathbf{j} + \frac{17}{5}\mathbf{k} \quad (3) \text{ or } (\frac{21}{5}, \frac{7}{5}, \frac{17}{5})
 \end{aligned}$$



$$\begin{aligned}
 \text{ii) } \text{proj}_{\vec{b}} \vec{p} &= \frac{-1(-2) + 6(5) - 1(-7)}{2^2 + 5^2 + 7^2} (-2\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}) \quad (1) \\
 &= \frac{1}{2}(-2\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}) \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } d &= |\text{proj}_{\vec{b}} \vec{p} - \vec{p}| \quad (1) \\
 &= |\frac{1}{2}(-2\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}) - (-\mathbf{i} + 6\mathbf{j} - \mathbf{k})| \\
 &= |0\mathbf{i} - \frac{7}{2}\mathbf{j} - \frac{5}{2}\mathbf{k}| \\
 &= \frac{\sqrt{74}}{2} \quad (2) \\
 &\text{or } \frac{\sqrt{37}}{\sqrt{2}}
 \end{aligned}$$



$$\begin{aligned}
 \text{a) } |z-6| &= |z+2i| \\
 \text{Let } x+iy &= z \\
 |x-6+iy| &= |x+i(y+2)| \\
 \sqrt{(x-6)^2 + y^2} &= \sqrt{x^2 + (y+2)^2} \quad (1) \\
 x^2 - 12x + 36 + y^2 &= x^2 + y^2 + 4y + 4 \\
 0 &= 12x + 4y - 32 \\
 0 &= 3x + y - 8 \quad (2)
 \end{aligned}$$

iii) Min. value of $|z|$ is $\perp$ distance of this line from $(0,0)$.
Need to find perpendicular line to P passing through origin.

$$\begin{aligned}
 \text{Gradient of } z \text{ locus is } -3 \\
 \therefore \text{Gradient of line through origin is } \frac{1}{3}. \therefore \text{Eq'n is } y = \frac{x}{3} \\
 \text{Solve simultaneously:} \\
 3x + \frac{x}{3} - 8 &= 0 \\
 \frac{10x}{3} &= 8 \\
 10x &= 24 \\
 x &= \frac{24}{10} = \frac{12}{5} \\
 \therefore y &= \frac{24}{30} = \frac{4}{5} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{min } |z| \text{ is } \sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{4}{5}\right)^2} \\
 = \frac{\sqrt{60}}{5} = \frac{2\sqrt{15}}{5} \quad (2)
 \end{aligned}$$



HSC Extension 2 Mathematics

209

Further Work with Vectors



Show that the line given by parametric vector equation $\underline{r} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ intersects the curve given by $\underline{v} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} t \\ t \\ t^2 \end{pmatrix}$ twice. 3



HSC Extension 2 Mathematics

210

Further Work with Vectors



The angle between the vector $x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ and the $x - z$ plane is:

- (A) $\sin^{-1} \left(\frac{y_1}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \right)$.
- (B) $\cos^{-1} \left(\frac{\sqrt{x_1^2 + z_1^2}}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \right)$.
- (C) $\cos^{-1} \left(\frac{y_1}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \right)$.
- (D) $\sin^{-1} \left(\frac{\sqrt{x_1^2 + z_1^2}}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \right)$.



HSC Extension 2 Mathematics

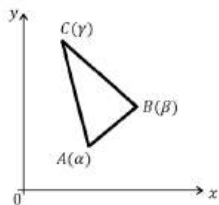
211

Further Work with Vectors



In the diagram, ABC is a triangle such that $AC = 2AB$ and $\angle BAC = \frac{\pi}{3}$.

The vertices A, B , and C are represented by the complex numbers α, β and γ respectively.



Which of the following is the correct expression of the complex number γ ?

- (A) $\gamma = (1 + i\sqrt{3})(\beta - \alpha) + \alpha$
- (B) $\gamma = (1 + i\sqrt{3})(\beta + \alpha) - \alpha$
- (C) $\gamma = \frac{1}{2}(1 + i\sqrt{3})(\beta - \alpha) + \alpha$
- (D) $\gamma = (1 + i\sqrt{3})(\alpha - \beta) + \alpha$



HSC Extension 2 Mathematics

212

Further Work with Vectors

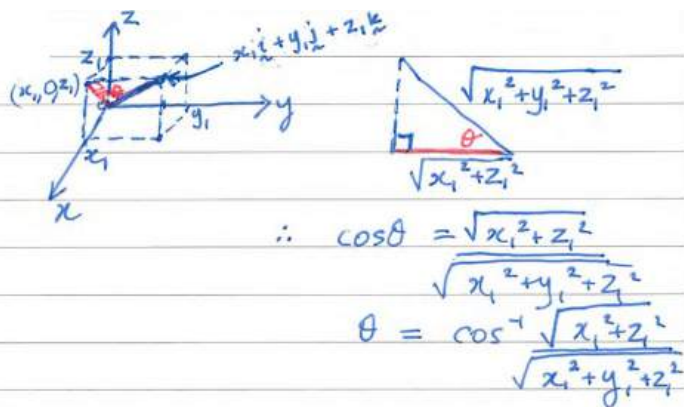


A, B and C are three collinear points with position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively. 2

Point B lies between A and C with $|\underline{BC}| = \frac{1}{2}|\underline{AB}|$.

Find $\underline{c}$ in terms of $\underline{a}$ and $\underline{b}$.





B



$$\underline{r} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} t \\ t \\ t \end{pmatrix}$$

Intersect if they have the same x, y, z components, i.e. if these eqs are consistent:

$$\textcircled{1} \quad 1 + \lambda = 2 + t$$

$$\textcircled{2} \quad -1 + \lambda = t$$

$$\textcircled{3} \quad 1 - \lambda = t$$

$$\textcircled{2} + \textcircled{3} \quad 0 = t + t$$

$$t(t+1) = 0$$

$$\text{i.e. } t=0, \lambda=1 \text{ or } t=-1, \lambda=0 \text{ (from } \textcircled{3}\text{)}.$$

Sub into $\textcircled{1}$:

$$\text{LHS} = 1 + 1$$

$$= 2$$

$$\text{RHS} = 2 + 0$$

$$= \text{LHS}$$

or

$$\text{LHS} = 1 + 0$$

$$\text{RHS} = 2 + (-1)$$

$$= \text{LHS}$$

$\therefore$ they intersect twice



If B divides AC into a ratio 2:1

then

$$|\vec{BC}| = \frac{1}{2} |\vec{AB}| \quad \dots$$

Now

$$\vec{BC} = \underline{c} - \underline{b} \quad \text{and} \quad \vec{AB} = \underline{b} - \underline{a}$$

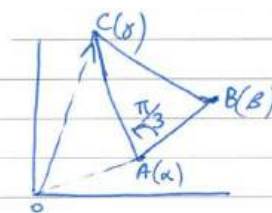
$$\therefore \underline{c} - \underline{b} = \frac{1}{2} (\underline{b} - \underline{a})$$

$$\underline{c} = \frac{1}{2} \underline{b} - \frac{1}{2} \underline{a} + \underline{b}$$

$$\underline{c} = \frac{3}{2} \underline{b} - \frac{1}{2} \underline{a}$$

or

$$= \frac{1}{2} (3\underline{b} - \underline{a})$$



By rotating vector $\vec{AB}$ by $\frac{\pi}{3}$ and multiplying its magnitude by 2 we obtain $\vec{AC}$.

$$\vec{AC} = \vec{AB} \times 2 \times \text{cis } \frac{\pi}{3}$$

$$= (\underline{b} - \underline{a}) \times 2 \times \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$= (\underline{b} - \underline{a}) \times 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$$

$$= (\underline{b} - \underline{a}) \times (1 + i\sqrt{3})$$

$$\text{but } \vec{AC} = \underline{y} - \underline{a}$$

$$\underline{y} - \underline{a} = (\underline{b} - \underline{a})(1 + i\sqrt{3})$$

$$\underline{y} = (\underline{b} - \underline{a})(1 + i\sqrt{3}) + \underline{a}$$

A



HSC Extension 2 Mathematics

213

Further Work with Vectors



The line l has equation $\frac{x-1}{k} = \frac{y-1}{3} = \frac{3-z}{6}$. Find the positive value of k for which the angle between line l and the x -axis is 120° . 3



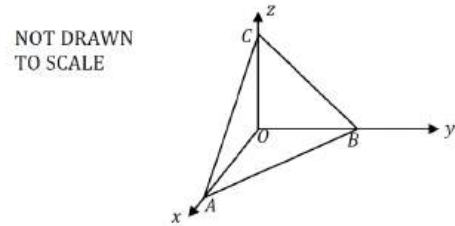
HSC Extension 2 Mathematics

214

Further Work with Vectors



The diagram $OABC$ below is a right triangular pyramid in which $|\overrightarrow{OA}| = 10$, $|\overrightarrow{OB}| = 20$ and $|\overrightarrow{OC}| = 10$, and $\angle AOB = \angle AOC = \angle BOC = 90^\circ$.



The unit vectors $\underline{i}$, $\underline{j}$ and $\underline{k}$ are parallel to $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively. The points R and S are the midpoints of AB and BC respectively.

- Determine the vectors $\overrightarrow{OR}$ and $\overrightarrow{OS}$ in terms of $\underline{i}$, $\underline{j}$ and $\underline{k}$. 2
- Calculate, to the nearest minute, the angle between $\overrightarrow{OR}$ and $\overrightarrow{OS}$. 2
- Find the exact length of the perpendicular from R to $\overrightarrow{OS}$. 2



HSC Extension 2 Mathematics

215

Further Work with Vectors



A sphere has a centre at $(3, -3, 4)$ and its radius is 6 units.

A line has equation $\underline{r} = \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}$.

- Write down the vector equation of the sphere. 1
- Determine whether the line is a tangent to the sphere, clearly justifying your conclusion. 3



HSC Extension 2 Mathematics

216

Further Work with Vectors



Lines l_1 and l_2 are given below, relative to a fixed origin O .

$$l_1: \underline{r}(t) = (-9\underline{i} + 10\underline{k}) + \gamma(2\underline{i} + \underline{j} - \underline{k})$$

$$l_2: \underline{r}(t) = (3\underline{i} + \underline{j} + 17\underline{k}) + \mu(3\underline{i} - \underline{j} + 5\underline{k})$$

where γ and μ are scalar parameters.

- Line l_1 meets line l_2 at the point A . What is the position vector of point A . 3
- Point B has position vector $5\underline{i} + 7\underline{j} + 3\underline{k}$. Does B lie on either line? 2



Further Work with Vectors Solution

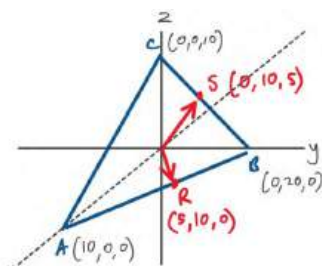
$$\text{i) } S = \left(\frac{0+10}{2}, \frac{0+20}{2}, \frac{10+0}{2} \right) \\ = (5, 10, 5) \\ \therefore \vec{OS} = 10\hat{j} + 5\hat{k}$$

$$R = \left(\frac{10+0}{2}, \frac{0+20}{2}, \frac{0+0}{2} \right) \\ = (5, 10, 0) \\ \therefore \vec{OR} = 5\hat{i} + 10\hat{j}$$

$$\text{ii) } \cos \theta = \frac{\begin{vmatrix} 0 & 10 & 5 \\ 5 & 10 & 0 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ 10 \\ 0 \end{vmatrix}}{\left| \begin{vmatrix} 0 & 10 & 5 \\ 5 & 10 & 0 \end{vmatrix} \right| \left| \begin{vmatrix} 5 \\ 10 \\ 0 \end{vmatrix} \right|} = \frac{100}{\sqrt{125} \times \sqrt{125}} = \frac{4}{5}$$

$$\therefore \theta = \cos^{-1} \frac{4}{5} = 36.8698^\circ \approx 36^\circ 52'$$

$$\text{iii) } \sin(\cos^{-1} \frac{4}{5}) = \frac{x}{\sqrt{125}} \\ \therefore \frac{3}{5} = \frac{x}{\sqrt{125}} \\ \therefore x = \sqrt{125} \times \frac{3}{5} = 3\sqrt{5}$$



Further Work with Vectors Solution

$$\frac{x-1}{k} = \frac{y-1}{3} = \frac{z-2}{6}$$

$$\text{L: } \frac{x-1}{k} = \frac{y-1}{3} = \frac{z-3}{-6}$$

$\therefore$ the direction vector of line L is: $\begin{pmatrix} k \\ 3 \\ -6 \end{pmatrix}$

Now the direction vector of the x-axis is: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Using the dot product,

$$\begin{pmatrix} k \\ 3 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \left| \begin{pmatrix} k \\ 3 \\ -6 \end{pmatrix} \right| \times \left| \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right| \cos 120^\circ$$

$$\cos 120^\circ = \frac{\begin{pmatrix} k \\ 3 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}{\sqrt{k^2 + 3^2 + (-6)^2} \times \sqrt{1^2}}$$

$$-\frac{1}{2} = \frac{k \times 1 + 3 \times 0 + (-6) \times 0}{\sqrt{k^2 + 9 + 36}}$$

$$-\frac{1}{2} = \frac{k}{\sqrt{k^2 + 45}}$$

$$-2k = \sqrt{k^2 + 45}$$

$$4k^2 = k^2 + 45$$

$$3k^2 = 45$$

$$k^2 = 15$$

$$k = \pm\sqrt{15} \text{ but } k > 0$$

$$\therefore k = \sqrt{15}$$



Further Work with Vectors Solution

$$\text{i) For } L_1: \vec{r}(t) = (-9\hat{i} + 10\hat{k}) + \lambda(2\hat{i} + \hat{j} - \hat{k}) \quad \text{--- (A)}$$

$$L_2: \vec{r}(t) = (3\hat{i} + \hat{j} + 17\hat{k}) + \mu(3\hat{i} - \hat{j} + 5\hat{k}) \quad \text{--- (B)}$$

If L_1 intersects the line L_2 then

$$\vec{r}_1 = \vec{r}_2$$

$$-9\hat{i} + 10\hat{k} + \lambda(2\hat{i} + \hat{j} - \hat{k}) = 3\hat{i} + \hat{j} + 17\hat{k} + \mu(3\hat{i} - \hat{j} + 5\hat{k})$$

$$\hat{i}: (-9 + 2\lambda) + \lambda\hat{j} + \hat{k}(10 - \lambda) = \hat{i}(3 + 3\mu) + \hat{j}(1 - \mu) + \hat{k}(17 + 5\mu)$$

Equating parts

$$\hat{i}: -9 + 2\lambda = 3 + 3\mu \quad \text{--- (1)}$$

$$\hat{j}: \lambda = 1 - \mu \quad \text{--- (2)}$$

$$\hat{k}: 10 - \lambda = 17 + 5\mu \quad \text{--- (3)}$$

$$2 \times (2) \quad 2\lambda = 2 - 2\mu \quad \text{--- (4)}$$

$$(1) - (4) \quad -9 = 1 + 5\mu \quad \text{--- (5)}$$

$$\mu = -2$$

$$\text{Sub } \mu = -2 \text{ into (2)}$$

$$\therefore \lambda = 1 - (-2)$$

$$\lambda = 3$$

$$\text{Sub in (A) = (B)}$$

$$\vec{r}_1: -9\hat{i} + 10\hat{k} + 3(2\hat{i} + \hat{j} - \hat{k})$$

$$= -3\hat{i} + 3\hat{j} + 7\hat{k}$$

$$\text{also } \vec{r}_2: 3\hat{i} + \hat{j} + 17\hat{k} - 2(3\hat{i} - \hat{j} + 5\hat{k})$$

$$= -3\hat{i} + 3\hat{j} + 7\hat{k}$$

$$\therefore \text{Position vector of A is:}$$

$$-3\hat{i} + 3\hat{j} + 7\hat{k} \text{ or } \begin{pmatrix} -3 \\ 3 \\ 7 \end{pmatrix}$$

$$\text{ii) For point B } \begin{pmatrix} 5 \\ 7 \\ 3 \end{pmatrix}$$

$$\text{to satisfy line } L_1: \begin{pmatrix} -9 \\ 0 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

$$\text{We need to find a scalar } \lambda, \text{ where}$$

$$\begin{pmatrix} 5 \\ 7 \\ 3 \end{pmatrix} = \begin{pmatrix} -9 \\ 0 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

$$\lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 3 \end{pmatrix} - \begin{pmatrix} -9 \\ 0 \\ 10 \end{pmatrix} = \begin{pmatrix} 14 \\ 7 \\ -7 \end{pmatrix} = 7 \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

$$\therefore \lambda = 7$$

$$\therefore B \text{ lies on line } L_1$$



Further Work with Vectors Solution

i) vector equation of a sphere is:

$$\left| \vec{r} - \begin{pmatrix} -3 \\ 3 \\ 4 \end{pmatrix} \right| = 6$$

ii) For the line $x = 1 + 2\lambda$

$$y = 5 - \lambda$$

$$z = 4 - \lambda$$

The cartesian equation of a sphere is

$$(x - 3)^2 + (y + 3)^2 + (z - 4)^2 = 36$$

$$\text{sub (i) } ((1 + 2\lambda) - 3)^2 + ((5 - \lambda) + 3)^2 + (4 - \lambda - 4)^2 = 36$$

$$(2\lambda - 2)^2 + (8 - \lambda)^2 + (-\lambda)^2 = 36$$

$$4\lambda^2 - 8\lambda + 4 + 64 - 16\lambda + \lambda^2 + \lambda^2 = 36$$

$$6\lambda^2 - 24\lambda + 36 = 0$$

$$\text{Now } \Delta = b^2 - 4ac$$

$$= 24^2 - 4 \times 6 \times 36$$

$$= -288 < 0$$

$\therefore$ This quadratic has no real solutions since $\Delta < 0$ so the line does not intersect the sphere.

$\therefore$ the line is NOT a tangent to the sphere.



HSC Extension 2 Mathematics

217

Further Work with Vectors

Consider the lines L_1 and L_2 , determined by the vector equations

$$L_1: \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad L_2: \mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Show that L_1 and L_2 intersect and are perpendicular, stating the coordinates of the point of intersection.



HSC Extension 2 Mathematics

218

Further Work with Vectors

Find the points of intersection between the sphere S and the line L , given below. 3

$$S: \left| \mathbf{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

$$L: \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

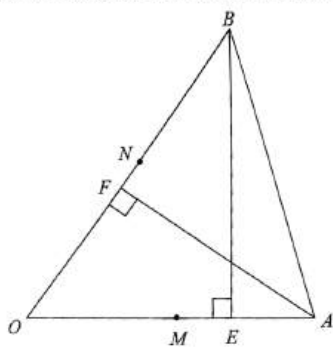


HSC Extension 2 Mathematics

219

Further Work with Vectors

In $\triangle OAB$, BE is the altitude from B to OA and AF is the altitude from A to OB . M and N are the midpoints of OA and OB respectively. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.



Not to scale

Use vector methods to show that $|OM||OE| = |ON||OF|$



3

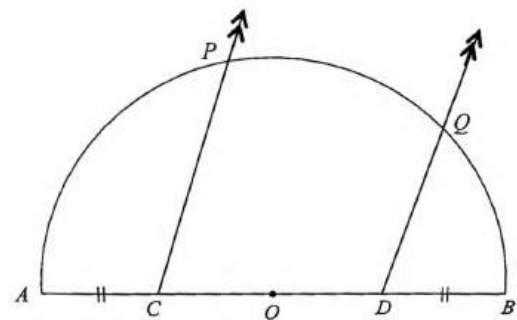


HSC Extension 2 Mathematics

220

Further Work with Vectors

A semi-circle is drawn on diameter AB . O is the midpoint of AB and the points C and D lie on AB such that $AC = BD$. Parallel lines are drawn through C and D , intersecting the semi-circle at P and Q respectively. $\overrightarrow{OC} = \mathbf{c}$ and $\overrightarrow{CP} = \mathbf{p}$.



- Explain why $\overrightarrow{DQ} = \lambda \mathbf{p}$ for some scalar $\lambda > 0$, then show that $(1 - \lambda) \mathbf{p} \cdot \mathbf{p} + 2 \mathbf{c} \cdot \mathbf{p} = 0$
- Hence show that $\angle CPQ = 90^\circ$.

3

2



Further Work with Vectors Solution

$$S: \left| \vec{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3 \quad \text{and} \quad L: \vec{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

POI when

$$\left| \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

$$\left| \begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} \right| = 3 \quad |\vec{a}|^2 = 9 \quad \checkmark$$

$$\begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} \cdot \begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} = 9$$

$$(1+\lambda)^2 + (1+\lambda)^2 + (-1+2\lambda)^2 = 9$$

$$1+2\lambda+\lambda^2 + 1+2\lambda+\lambda^2 + 1-4\lambda+4\lambda^2 = 9$$

$$6\lambda^2 = 6$$

$$\lambda = \pm 1 \quad \checkmark$$

Sub $\lambda = 1$ in L : $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$
 $= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

Sub $\lambda = -1$ in L : $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$
 $= \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$

✓ Both POIs



Further Work with Vectors Solution

$$L_1: \vec{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad L_2: \vec{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 2 \times 1 + -1 \times 1 + 1 \times -1 = 0$$

$$\therefore L_1 \perp L_2 \quad \checkmark$$

Then, POI when $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

$$3+2\lambda = -1+\mu \quad \text{--- (1)}$$

$$2-\lambda = 1+\mu \quad \text{--- (2)}$$

$$-1+\lambda = -\mu \quad \text{--- (3)}$$

$$\textcircled{1} - \textcircled{2}: 1+3\lambda = -2$$

$$\lambda = -1 \quad \text{--- (4)}$$

Sub (4) in (3): $2 - (-1) = 1 + \mu$

$$\mu = 2 \quad \checkmark$$

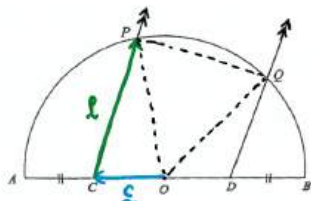
Test $\lambda = -1, \mu = 2$ in (1): LHS = $-1 - 1 = -2$

$$\text{RHS} = -2$$

$\therefore$ POI at $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} - 1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$
 $= \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \quad \checkmark$



Further Work with Vectors Solution



$$(i) \vec{OA} = \lambda \vec{CP} \quad (\text{OA} \parallel \text{CP})$$

$$\therefore \vec{OA} = \lambda \vec{r} \quad \text{for } \lambda > 0 \text{ scalar}$$

$$\text{Now, } \vec{OP} = \vec{OC} + \vec{CP}$$

$$= \vec{r} + \vec{r}$$

$$\vec{OA} = \vec{OC} + \vec{OP}$$

$$= -\vec{r} + \lambda \vec{r}$$

$$\text{Then, } |\vec{OA}| = |\vec{OP}| \quad (\text{equal radii of circle})$$

$$(\lambda + \vec{r}) \cdot (\lambda + \vec{r}) = (-\vec{r} + \lambda \vec{r}) \cdot (-\vec{r} + \lambda \vec{r})$$

$$\cancel{\lambda} \cdot \cancel{\lambda} + 2\vec{r} \cdot \vec{r} + \vec{r} \cdot \vec{r} = \cancel{\lambda} \cdot \cancel{\lambda} - 2\vec{r} \cdot \vec{r} + \lambda^2 \vec{r} \cdot \vec{r}$$

$$\vec{r} \cdot \vec{r} (1 - \lambda^2) = -2\vec{r} \cdot \vec{r} (1 + \lambda)$$

$$\vec{r} \cdot \vec{r} (1 - \lambda)(1 + \lambda) = -2\vec{r} \cdot \vec{r} (1 + \lambda)$$

$$\therefore (1 - \lambda) \vec{r} \cdot \vec{r} + 2\vec{r} \cdot \vec{r} = 0$$

$$(ii) \vec{PA} = \vec{PC} + \vec{CB} + \vec{BA}$$

$$= -\vec{r} - 2\vec{r} + \lambda \vec{r}$$

$$= (\lambda - 1) \vec{r} - 2\vec{r}$$

$$\text{Now, } \vec{CP} \cdot \vec{PA} = \vec{r} \cdot ((\lambda - 1) \vec{r} - 2\vec{r})$$

$$= (\lambda - 1) \vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{r}$$

$$= -[(1 - \lambda) \vec{r} \cdot \vec{r} + 2\vec{r} \cdot \vec{r}] = 0$$

$$\therefore \text{CP} \perp \text{PA}$$

$$\therefore \angle \text{CPA} = 90^\circ$$

from (i)



Further Work with Vectors Solution

$$\vec{OA} = \frac{1}{2} \vec{OB} = \frac{1}{2} \vec{a}$$

$$\vec{OE} = \lambda \vec{OA} = \lambda \vec{a} \quad \lambda \text{ scalar}$$

$$\vec{ON} = \frac{1}{2} \vec{OB} = \frac{1}{2} \vec{b}$$

$$\vec{OF} = \mu \vec{OB} = \mu \vec{b} \quad \mu \text{ scalar} \quad \checkmark$$

$$\text{Now, } |\text{OM}| |\text{OE}| = \frac{\Delta}{2} |\vec{a}|^2$$

$$|\text{OM}| |\text{OF}| = \frac{\Delta}{2} |\vec{b}|^2 \quad \checkmark$$

$$\text{Also, } \text{AF} \perp \text{OB, so } \vec{AF} \cdot \vec{OB} = 0$$

$$(\mu \vec{b} - \frac{1}{2} \vec{a}) \cdot \vec{b} = 0$$

$$\mu \vec{b} \cdot \vec{b} - \frac{1}{2} \vec{a} \cdot \vec{b} = 0$$

$$\vec{a} \cdot \vec{b} = \mu \vec{b} \cdot \vec{b} = \mu |\vec{b}|^2$$

$$\text{Similarly, } \text{BE} \perp \text{OA, so } \vec{BE} \cdot \vec{OA} = 0$$

$$(\vec{b} - \lambda \vec{a}) \cdot \vec{a} = 0$$

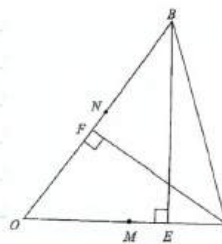
$$\vec{a} \cdot \vec{b} - \lambda \vec{a} \cdot \vec{a} = 0$$

$$\vec{a} \cdot \vec{b} = \lambda \vec{a} \cdot \vec{a} = \lambda |\vec{a}|^2$$

$$\text{Then } \vec{a} \cdot \vec{b} = \lambda |\vec{a}|^2 = \mu |\vec{b}|^2 \quad \checkmark$$

$$\frac{\Delta}{2} |\vec{a}|^2 = \frac{\Delta}{2} |\vec{b}|^2$$

$$\therefore |\text{OM}| |\text{OE}| = |\text{OM}| |\text{OF}|$$



HSC Extension 2 Mathematics

221

Further Work with Vectors



Relative to a fixed origin O , the point A has position vector $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$, the point B has position vector $4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, and the point C has position vector $a\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$, where a is a constant and $a < 0$.

D is the point such that $\overrightarrow{AB} = \overrightarrow{BD}$.

- (i) Find the position vector of D . 2
- (ii) If $|\overrightarrow{AC}| = 4$, find the value of a . 3



HSC Extension 2 Mathematics

222

Further Work with Vectors



A sphere is represented by the equation

$$x^2 + 2x + y^2 - 5y + z^2 - 4z + k = 0.$$

- (i) Determine the centre, C , and radius of the sphere. 2
- (ii) It is given that the $x-z$ plane is tangential to the sphere. Determine the value of k . 1



HSC Extension 2 Mathematics

223

Further Work with Vectors



Consider points $A(1, 3, 1)$, $B(2, 5, -1)$ and $C(7, 5, -1)$.
Line L passes through point B and bisects $\angle ABC$.

Find the vector equation of line L .

2



HSC Extension 2 Mathematics

224

Further Work with Vectors



With respect to a fixed origin O , the lines l_1 and l_2 are given by the equations:

$$l_1: r = 9\mathbf{i} + 13\mathbf{j} - 3\mathbf{k} + \lambda(\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) \text{ and } l_2: r = 2\mathbf{i} - \mathbf{j} + \mathbf{k} + \mu(2\mathbf{i} + \mathbf{j} + \mathbf{k})$$

where λ and μ are scalar parameters.

- (i) Given that l_1 and l_2 meet, find the position vector of their point of intersection. 3
- (ii) Find the acute angle between l_1 and l_2 , correct to the nearest tenth of a degree. 2
- (iii) Given that the point A has position vector $4\mathbf{i} + 16\mathbf{j} - 3\mathbf{k}$, and that the point $P(x_1, y_1, z_1)$ lies on l_1 such that AP is perpendicular to l_1 , find the exact coordinates of P . 3



Further Work with Vectors Solution

(i)

$$x^2 + 2x + 1 + y^2 - 5y + \frac{25}{4} + z^2 - 4z + 4 = \frac{45}{4} - k$$

$$(x+1)^2 + \left(y - \frac{5}{2}\right)^2 + (z-2)^2 = \frac{45-4k}{4}$$

$$\therefore \text{Centre} = \left(-1, \frac{5}{2}, 2\right) \text{ Radius} = \frac{\sqrt{45-4k}}{2}$$

(ii) Distance from the x - z plane is the y -value.

$$\frac{\sqrt{45-4k}}{2} = \frac{5}{2}$$

$$45 - 4k = 25 \rightarrow k = 5$$



Further Work with Vectors Solution

(i)

$$\overrightarrow{AB} = \overrightarrow{BD} = \begin{pmatrix} 2 \\ -5 \\ 7 \end{pmatrix} \quad \overrightarrow{AD} = 2\overrightarrow{AB} = \begin{pmatrix} 4 \\ -10 \\ 14 \end{pmatrix}$$

$$\text{Hence position vector for } D: \begin{pmatrix} 2+4 \\ 3-10 \\ -4+10 \end{pmatrix} = \begin{pmatrix} 6 \\ -7 \\ 10 \end{pmatrix}$$

(ii)

$$\overrightarrow{AC} = \begin{pmatrix} a-2 \\ 2 \\ 2 \end{pmatrix}$$

$$|\overrightarrow{AC}| = 4 \quad \therefore (a-2)^2 + 2^2 + 2^2 = 16$$

$$a = 2 - 2\sqrt{2} \text{ since } a \text{ is negative.}$$



Further Work with Vectors Solution

(i) Solving l_1 and l_2 simultaneously,

$$\begin{pmatrix} 9 \\ 13 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

so, $9 + \lambda = 2 + 2\mu$ and $13 + 4\lambda = -1 + \mu$

$$\lambda = 2\mu - 7 \quad \dots [1] \quad \mu = 4\lambda + 14 \quad \dots [2]$$

sub [1] $\rightarrow$ [2]

$$\mu = 4(2\mu - 7) + 14 \quad \mu = 2$$

$$-7\mu = -14 \quad \text{and } \lambda = 2(2) - 7 = -3$$

$$\therefore r = 9i + 13j - 3k + \lambda(i + 4j - 2k) = 9i + 13j - 3k - 3(i + 4j - 2k) = 6i + j + 3k$$

(ii) l_1 has direction vector $a = i + 4j - 2k$ l_2 has direction vector $b = 2i + j + k$

$$\cos \theta = \frac{a \cdot b}{|a| |b|} = \frac{1 \times 2 + 4 \times 1 + (-2) \times 1}{\sqrt{1^2 + 4^2 + (-2)^2} \times \sqrt{2^2 + 1^2 + 1^2}} = \frac{4}{\sqrt{21} \times \sqrt{6}}$$

$$\theta = \cos^{-1} \left(\frac{4}{\sqrt{126}} \right) = 69.1^\circ$$

(iii) P lies on l_1 , so

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 9 \\ 13 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$$

ie $x_1 = 9 + \lambda$

$$y_1 = 13 + 4\lambda$$

$$z_1 = -3 - 2\lambda$$

$$\text{Now, } \overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} - \begin{pmatrix} 9 \\ 16 \\ -3 \end{pmatrix} = \begin{pmatrix} 9 + \lambda - 9 \\ 13 + 4\lambda - 16 \\ -3 - 2\lambda + 3 \end{pmatrix} = \begin{pmatrix} \lambda \\ -3 + 4\lambda \\ -2\lambda \end{pmatrix}$$

noting that l_1 has direction vector $i + 4j - 2k$,

$$l_1 \perp \overrightarrow{AP}$$

$$\text{then, } (i + 4j - 2k) \cdot \overrightarrow{AP} = 0$$

$$1(\lambda) + 4(-3 + 4\lambda) - 2(-2\lambda) = 0$$

$$-7 + 21\lambda = 0 \quad \lambda = \frac{1}{3}$$

Hence, P is $\left(9 + \frac{1}{3}, 13 + 4\left(\frac{1}{3}\right), -3 - 2\left(\frac{1}{3}\right)\right)$ from

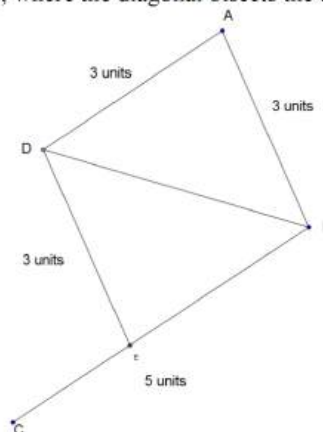
$$= \left(\frac{28}{3}, \frac{43}{3}, -\frac{11}{3}\right)$$



Further Work with Vectors Solution

$$|\overrightarrow{BA}| = 3 \quad |\overrightarrow{BC}| = 5$$

Consider the diagram below that utilises the property of a rhombus, where the diagonal bisects the angles it passes through.

For BD to bisect $\angle ABC$ we have the following vector sum.

$$\overrightarrow{BD} = \overrightarrow{BA} + \frac{3}{5} \overrightarrow{BC}$$

$$= \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} + \frac{3}{5} \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix}$$

This is a direction vector for line L . Line L also passes through point B .

$$L = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$



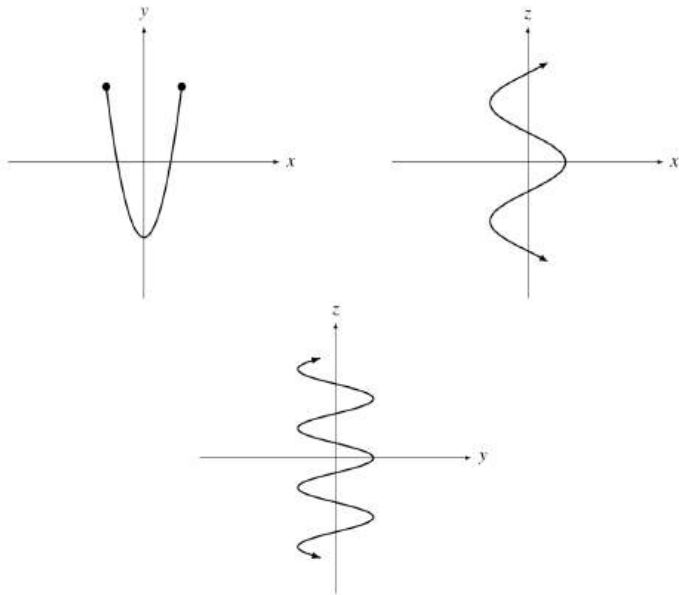
HSC Extension 2 Mathematics

225

Further Work with Vectors



Let $\underline{r}(t)$ be a curve in 3D space. The following diagrams show projections of $\underline{r}(t)$ onto the xy , xz and yz planes. The diagrams are NOT to scale.



Which of the following is the correct vector representation of $\underline{r}(t)$?

- A. $\underline{r}(t) = (2 \sin t)\underline{i} + (4 \cos 2t)\underline{j} + (t)\underline{k}$
- B. $\underline{r}(t) = (2 \sin t)\underline{i} + (4 \sin 2t)\underline{j} + (t)\underline{k}$
- C. $\underline{r}(t) = (2 \cos t)\underline{i} + (4 \cos 2t)\underline{j} + (t)\underline{k}$
- D. $\underline{r}(t) = (2 \cos t)\underline{i} + (4 \sin 2t)\underline{j} + (\sqrt{t})\underline{k}$



HSC Extension 2 Mathematics

226

Further Work with Vectors



Find the cartesian equation of the vector function $\underline{r}(t) = \begin{bmatrix} \frac{1}{2}(\cos t - \sin t) \\ \cos^3 t - \sin^3 t \end{bmatrix}$. 4



HSC Extension 2 Mathematics

227

Further Work with Vectors



Three points A , B and C have position vectors $-2\underline{a} + 3\underline{b} + 5\underline{c}$, $\underline{a} + 2\underline{b} + 3\underline{c}$ and $7\underline{a} - \underline{c}$ respectively. The point B lies between A and C . Show that A , B and C are collinear. 2



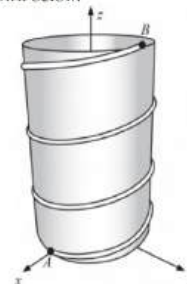
HSC Extension 2 Mathematics

228

Further Work with Vectors



The stairs of a cylindrical shaped tower spiral upwards from the ground to an observation deck at point B , as shown below.



The path begins at point A on the ground along the x -axis and finishes at B . The time taken in seconds for Hannah to walk along the spiral path starting at A is presented by t . It takes Hannah 70π seconds to reach point B . Her position on this path can be represented by

$$\underline{r}(t) = \begin{bmatrix} 15 \cos(0.5t) \\ 15 \sin(0.5t) \\ 0.3t \end{bmatrix} \text{ metres.}$$

- (i) Determine the height of the observation deck above the ground correct to 2 decimal places. 1

Hannah walks one loop around the tower and ends up directly above point A .

- (ii) At what time and height does this occur? 1

- (iii) Find Hannah's speed at this point. 2



$$x = \frac{1}{2}(\cos t - \sin t),$$

$$2x = \cos t - \sin t$$

$$y = \cos^3 t - \sin^3 t$$

$$= (\cos t - \sin t)(\cos^2 t + \cos t \sin t + \sin^2 t)$$

$$= 2x(1 + \cos t \sin t)$$

Now,

$$4x^2 = \cos^2 t - 2\cos t \sin t + \sin^2 t$$

$$= 1 - 2\cos t \sin t$$

$$\therefore \cos t \sin t = \frac{1}{2}(1 - 4x^2)$$

$$y = 2x\left(1 + \frac{1}{2}(1 - 4x^2)\right)$$

$$= 2x + x(1 - 4x^2)$$

$$= 3x - 4x^3$$

Using auxiliary angle:

$$x = \frac{\sqrt{2}}{2} \cos(t - \alpha)$$

$$\therefore -\frac{\sqrt{2}}{2} \leq x \leq \frac{\sqrt{2}}{2}$$

$$\therefore y = 3x - 4x^3, -\frac{\sqrt{2}}{2} \leq x \leq \frac{\sqrt{2}}{2}$$



Clearly not (D) since $z = \sqrt{t} \geq 0$ but $z \in \mathbb{R}$.

Clearly not (A) or (B) since the x - z plane shows $2\cos(z) = x$, not $2\sin(z) = x$.



i) height = $0.3 \times 70\pi = 21\pi$
 $= 65.97344... = 65.97\text{m (2dp)}$

ii) when $t=0$, $A = \begin{pmatrix} 15 \\ 0 \end{pmatrix}$

$\therefore$ for point above A:

$$15 \cos(0.5t) = 15$$

$$\cos(0.5t) = 1$$

$$0.5t = 0, 2\pi, 4\pi, \dots$$

$$\therefore t = 0, 4\pi, 8\pi, \dots$$

$$\therefore t = 4\pi$$

when $t = 4\pi$, height = $0.3 \times 4\pi = 1.2\pi$
 $= 3.7699... = 3.77\text{m (2dp)}$

iii) $\mathbf{r}'(t) = \begin{bmatrix} -7.5 \sin(0.5t) \\ 7.5 \cos(0.5t) \\ 0.3 \end{bmatrix}$

at $t = 4\pi$

$$\text{Speed} = \sqrt{(-7.5 \sin(0.5 \times 4\pi))^2 + (7.5 \cos(0.5 \times 4\pi))^2 + 0.3^2}$$

$$= 7.505997... = 7.51\text{m/s (2dp)}$$



$$\vec{AB} = 3\mathbf{a} - \mathbf{b} - 2\mathbf{c}$$

$$\vec{BC} = 6\mathbf{a} - 2\mathbf{b} - 4\mathbf{c}$$

$$\vec{AC} = 9\mathbf{a} - 3\mathbf{b} - 6\mathbf{c}$$

$$\vec{BC} = 2\vec{AB} \quad \text{or} \quad \vec{AC} = 3\vec{AB}$$

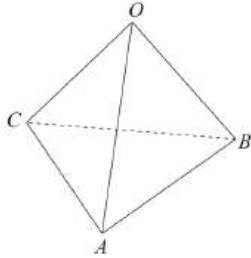
$\therefore$ the points are collinear as $\vec{BC} \parallel \vec{AB}$ and B is a common point



HSC Extension 2 Mathematics

229

Further Work with Vectors



The diagram above shows the tetrahedron $OABC$. That is, $OABC$ is a triangular pyramid where each face is an equilateral triangle. Each edge of the pyramid has length 1 unit.

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, and $\overrightarrow{OC} = \underline{c}$.

What is the value of $|\text{proj}_{\underline{c}}(\text{proj}_{\underline{c}}(\text{proj}_{\underline{b}}\underline{a}))|$?

- (A) $\frac{1}{8}$
- (B) $\frac{1}{4}$
- (C) 1
- (D) 4



HSC Extension 2 Mathematics

230

Further Work with Vectors



Consider the line $\underline{r} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. The point $A(x, y)$ lies on the line, whereas the point $B(-1, 3)$ does not.

If O is the origin, find the coordinates of A if $\overrightarrow{OA} \perp \overrightarrow{OB}$.



HSC Extension 2 Mathematics

231

Further Work with Vectors



For two vectors $\underline{a}$ and $\underline{b}$, it is known that $|\underline{a}| = 5$, $|\underline{b}| = 3$, and $\underline{a} \cdot \underline{b} = 9$.

- (i) Show that $\underline{b} \cdot (\underline{a} - \underline{b}) = 0$. 1
- (ii) Find $|\underline{a} - \underline{b}|$. 2
- (iii) If $\underline{a} = \overrightarrow{OA}$ and $\underline{b} = \overrightarrow{OB}$, find the area of $\triangle OAB$. 1



HSC Extension 2 Mathematics

232

Further Work with Vectors

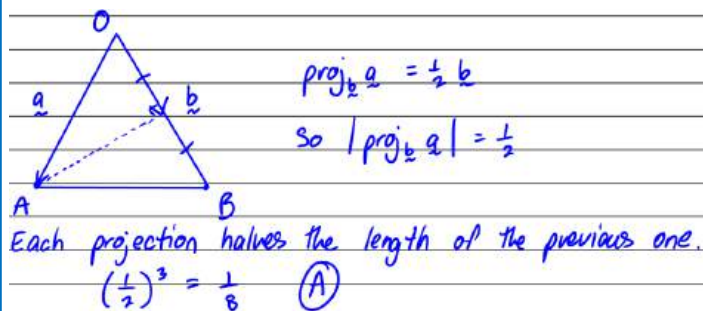


Three points $A(-2, 1, 7)$, $B(-1, 3, 5)$, and $C(-4, 7, 3)$ lie in three-dimensional space. The points A and B lie on the line l .

- (i) Express $\overrightarrow{AB}$ and $\overrightarrow{AC}$ as column vectors. 1
- (ii) Find the shortest distance from C to the line l . 3



$$\begin{aligned} \vec{r} &= \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 4 + \lambda \\ -2 + 2\lambda \end{bmatrix} \\ \therefore \vec{OA} &= \begin{bmatrix} 4 + \lambda \\ -2 + 2\lambda \end{bmatrix} \\ \vec{OA} \cdot \vec{OB} &= \begin{bmatrix} 4 + \lambda \\ -2 + 2\lambda \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \end{bmatrix} \\ &= -4 - \lambda - 6 + 6\lambda \\ &= -10 + 5\lambda \\ \therefore \vec{OA} \cdot \vec{OB} &= 0 \text{ when } \lambda = 2 \\ \therefore \vec{OA} &= \begin{bmatrix} 4 + 2 \\ -2 + 2 \times 2 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ 2 \end{bmatrix} \\ \therefore A &\text{ is the point } (6, 2). \end{aligned}$$



(i) $A(-2, 1, 7)$, $B(-1, 3, 5)$, $C(-4, 7, 3)$

$$\vec{AB} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \quad \vec{AC} = \begin{bmatrix} -2 \\ 6 \\ -4 \end{bmatrix}$$

(ii)

$$\begin{aligned} \text{proj}_{\vec{AB}} \vec{AC} &= \frac{\vec{AB} \cdot \vec{AC}}{\vec{AB} \cdot \vec{AB}} \vec{AB} \\ &= \frac{-2 + 12 + 8}{1 + 4 + 4} \vec{AB} \\ &= \frac{18}{9} \vec{AB} \\ &= 2 \vec{AB} \\ &= \begin{bmatrix} 2 \\ 4 \\ -4 \end{bmatrix} \\ \therefore \vec{d} &= \vec{AB} - \text{proj}_{\vec{AB}} \vec{AC} \\ &= \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ -4 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix} \\ \text{and } |\vec{d}| &= \sqrt{(-1)^2 + (-2)^2 + 2^2} \\ &= \sqrt{1 + 4 + 4} \\ &= \sqrt{9} \\ &= 3 \text{ units} \end{aligned}$$



(i) $\vec{b} \cdot (\vec{a} - \vec{b}) = \vec{b} \cdot \vec{a} - \vec{b} \cdot \vec{b}$

$$\begin{aligned} &= \vec{b} \cdot \vec{a} - |\vec{b}|^2 \\ &= 9 - 3^2 = 0 \end{aligned}$$

(ii) $|\vec{a} - \vec{b}|^2 = (\vec{a} - \vec{b}) \cdot (\vec{a} - \vec{b})$

$$\begin{aligned} &= \vec{a} \cdot \vec{a} - 2\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} \\ &= |\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \\ &= 5^2 - 2 \times 9 + 9 = 16 \\ \therefore |\vec{a} - \vec{b}| &= \sqrt{16} = 4 \end{aligned}$$

(iii) From (i), $\vec{OB} \perp \vec{BA}$

$$\begin{aligned} \therefore \text{Area} &= \frac{1}{2} \times |\vec{OB}| \times |\vec{BA}| \\ &= \frac{1}{2} \times 3 \times 4 \\ &= 6 \text{ units}^2 \end{aligned}$$



HSC Extension 2 Mathematics

233

Further Work with Vectors



Consider the sphere S , centred at point $C(2, -1, 0)$ with radius $\sqrt{29}$. The line L with parametric equations $\begin{cases} x = \lambda + 1 \\ y = \lambda \\ z = 2\lambda + 3 \end{cases}$ intersects the surface of the sphere S at points P and Q .

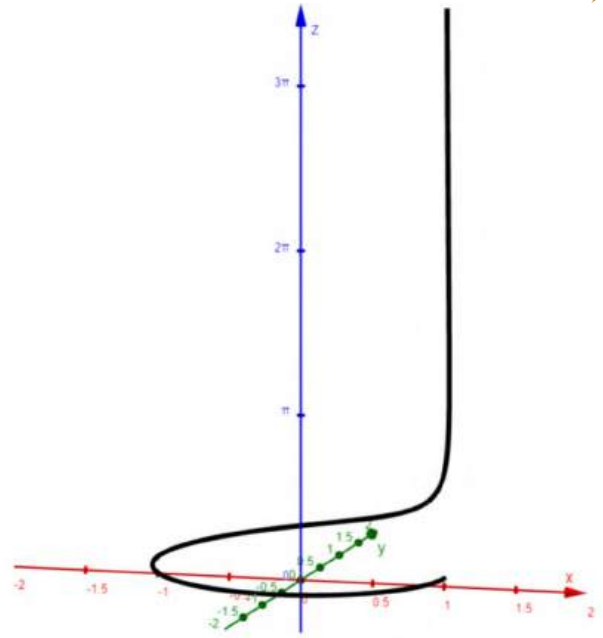
- Find the coordinates of the points P and Q . 2
- A line parallel to line L is a tangent touching the sphere S at a single point, R . 2
Find the possible coordinates of point R .



HSC Extension 2 Mathematics

234

Further Work with Vectors



Which of the following parametric equations best describes the graph above.

- $x = \cos t, y = \sin t, z = t$
- $x = \cos t, y = \sin t, z = \frac{1}{t}$
- $x = \cos t, y = t, z = \sin t$
- $x = \cos t, y = \frac{1}{t}, z = \sin t$



HSC Extension 2 Mathematics

235

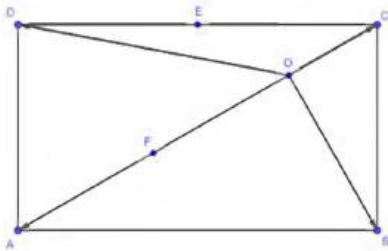
Further Work with Vectors



In the diagram below, O is the origin and points A, B, C and D have position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ and $\mathbf{d}$ respectively.

E is the midpoint of CD and F is the midpoint of OA .

$ABCD$ is a rectangle such that O lies on diagonal AC , and OB is perpendicular to AC .



- Show that $|\mathbf{b}|^2 + \mathbf{a} \cdot \mathbf{c} = 0$. 2
- Hence, or otherwise, show that $\angle BFE = \frac{\pi}{2}$. 3



HSC Extension 2 Mathematics

236

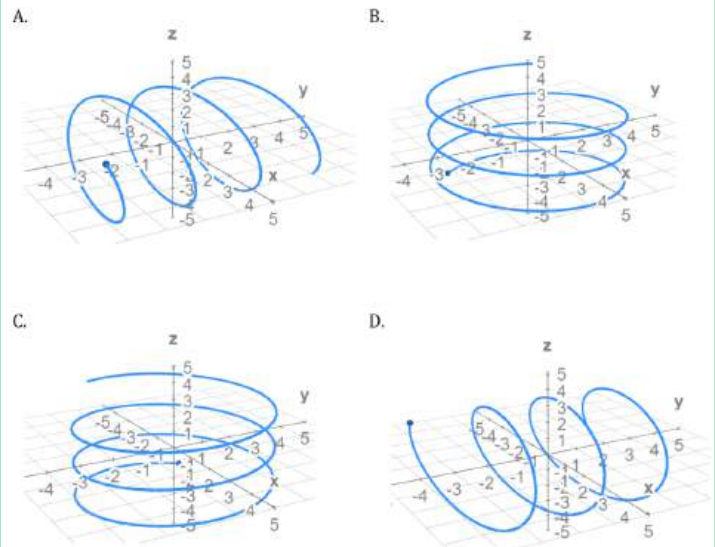
Further Work with Vectors



Which diagram best shows the curve described by the position vector:

$$\mathbf{r}(t) = 3 \cos(t) \mathbf{i} + \frac{t}{3} \mathbf{j} + 3 \sin(t) \mathbf{k}$$

for $t \in [-10, 10]$?



at $t = 0$ we see
 $x = 1$, $y = 0$ &
 z is undefined

B. $x = \cos t$, $y = \sin t$, $z = \frac{1}{t}$



$$\begin{aligned} \text{(i)} \quad x &= \lambda + 1 & |y - c| &= \sqrt{29} \\ y &= \lambda & \left| \begin{pmatrix} \lambda + 1 \\ \lambda \\ 2\lambda + 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \right| &= \sqrt{29} \\ z &= 2\lambda + 3 & \left| \begin{pmatrix} \lambda - 1 \\ \lambda + 1 \\ 2\lambda + 3 \end{pmatrix} \right| &= \sqrt{29} \\ & & (\lambda - 1)^2 + (\lambda + 1)^2 + (2\lambda + 3)^2 &= 29 \\ & & 6\lambda^2 + 12\lambda - 18 &= 0 \\ & & \lambda^2 + 2\lambda - 3 &= 0 \\ & & \lambda &= -3, \lambda = 1 \\ & & \therefore P: (-2, -3, -3), Q: (2, 1, 5) \end{aligned}$$

(ii) Since the line is parallel to L and tangent to the sphere at R , CR will be perpendicular to L and bisect PQ . Let M be the mid-point of PQ and, hence, the intersection point of PQ and CR . Therefore:

$$P: (-2, -3, -3), Q: (2, 1, 5), C: (2, -1, 0)$$

$$M: (0, -1, 1)$$

$$\overrightarrow{CM} = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

As $\overrightarrow{CR}$ is an extension of $\overrightarrow{CM}$ with a length of $\sqrt{29}$:

$$\overrightarrow{CR} = \sqrt{\frac{29}{5}} \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

$$\overrightarrow{OR} = \sqrt{\frac{29}{5}} \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$$

$$R: \left(2 - 2\sqrt{\frac{29}{5}}, -1, \sqrt{\frac{29}{5}} \right)$$



In the XZ plane we have a circle with radius 3 so either option A. or D. is correct. When $t = -10$, $r(t) = 3 \cos(-10)\mathbf{i} - \frac{10}{3}\mathbf{j} + 0\mathbf{k}$, but $3 \cos(-10) < 0$ so D is correct.

D



$$\text{(i)} \quad \overrightarrow{AB} \cdot \overrightarrow{BC} = 0 \quad (\text{sides of rectangle perpendicular})$$

$$(b - a)(c - b) = 0$$

$$b \cdot c - b \cdot b - a \cdot c + a \cdot b = 0$$

$$\therefore -|b|^2 - a \cdot c = 0 \quad (\text{as } a \perp b \text{ \& } b \perp c)$$

$$(\times -1) \quad \therefore |b|^2 + a \cdot c = 0$$

(ii)

$$\begin{aligned} \overrightarrow{BF} &= \overrightarrow{BO} + \overrightarrow{OF} \\ &= -\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{a} \\ &= \frac{1}{2}\mathbf{a} - \mathbf{b} \end{aligned}$$

$$\begin{aligned} \overrightarrow{FE} &= \overrightarrow{FO} + \overrightarrow{OE} + \overrightarrow{CE} \\ &= -\frac{1}{2}\mathbf{a} + \mathbf{c} + \frac{1}{2}(\mathbf{a} - \mathbf{b}) \end{aligned}$$

$$\boxed{\sqrt{\frac{BF}{FE}}}$$

$$\begin{aligned} &= -\frac{1}{2}\mathbf{a} + \mathbf{c} + \frac{1}{2}\mathbf{a} - \frac{1}{2}\mathbf{b} \\ &= \mathbf{c} - \frac{1}{2}\mathbf{b} \end{aligned}$$

$$\begin{aligned} \therefore \overrightarrow{BF} \cdot \overrightarrow{FE} &= \left(\frac{1}{2}\mathbf{a} - \mathbf{b}\right) \cdot \left(\mathbf{c} - \frac{1}{2}\mathbf{b}\right) \quad \boxed{\checkmark \text{ dot product}} \\ &= \frac{1}{2}\mathbf{a} \cdot \mathbf{c} - \frac{1}{2}\mathbf{a} \cdot \mathbf{b} - \mathbf{b} \cdot \mathbf{c} + \frac{1}{2}\mathbf{b} \cdot \mathbf{b} \quad \boxed{\checkmark \text{ result from (i)}} \\ &= \frac{1}{2}(\mathbf{a} \cdot \mathbf{c} + |b|^2) \\ &= 0 \quad \therefore \overrightarrow{BF} \perp \overrightarrow{FE} \end{aligned}$$



HSC Extension 2 Mathematics

237

Further Work with Vectors



The line l_1 passes through the points $A(2, -1, 4)$ and $B(4, -3, 2)$.

(i) Show that a vector equation of line l_1 is $\mathbf{r} = \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$. 2

(ii) The equation of line l_2 is given by $\frac{x+3}{k} = \frac{4-y}{6} = z-1$. 3

Find the value of k for which l_1 and l_2 intersect.



HSC Extension 2 Mathematics

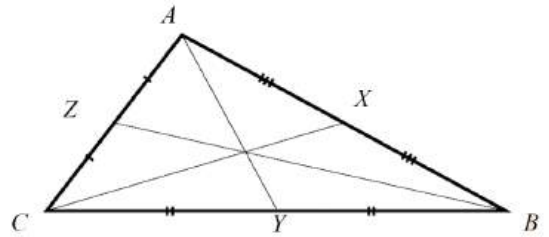
238

Further Work with Vectors



The diagram shows a triangle ABC .

The points X , Y and Z bisect the intervals AB , BC and CA respectively.



Show that $\overrightarrow{AY} + \overrightarrow{BZ} + \overrightarrow{CX} = \mathbf{0}$



HSC Extension 2 Mathematics

239

Further Work with Vectors



Relative to a fixed origin O , the vector equations of two lines l_1 and l_2 are:

$$l_1: \mathbf{r} = \begin{pmatrix} 9 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ -3 \\ 5 \end{pmatrix} \quad \text{and} \quad l_2: \mathbf{r} = \begin{pmatrix} -16 \\ \alpha \\ 10 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -4 \\ 9 \end{pmatrix}$$

where α is a constant and λ and μ are scalars.

(i) Given that the two lines intersect, find the value of α , and find the position vector of the point of intersection A . 3

(ii) Show that the acute angle between l_1 and l_2 is 60° . 1

(iii) If B is a point on l_1 such that AB is 10 units, find the perpendicular distance from B to the line l_2 . 1



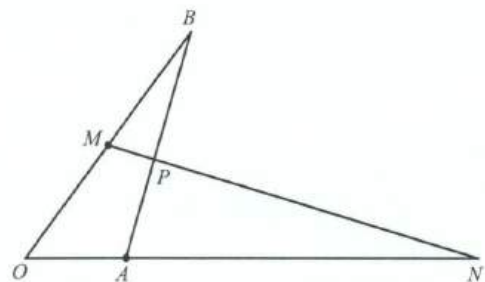
HSC Extension 2 Mathematics

240

Further Work with Vectors



In the diagram below, ON , OB , AB and MN are all straight lines.



M is the midpoint of OB , and $OA:AN = 1:4$.

$\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OB} = 2\mathbf{b}$.

Let $\overrightarrow{AP} = \lambda \overrightarrow{AB}$, where λ is a scalar.

(i) Show that $\overrightarrow{OP} = 2(1-\lambda)\mathbf{a} + 2\lambda\mathbf{b}$. 1

(ii) If $\overrightarrow{MP} = \mu \overrightarrow{MN}$, where μ is a scalar, find another expression for $\overrightarrow{OP}$ in terms of μ , $\mathbf{a}$ and $\mathbf{b}$. 2

(iii) Hence find the ratio in which P divides AB . 2



Further Work with Vectors Solution

$$\overrightarrow{CA} + \overrightarrow{AB} + \overrightarrow{BC} = 0 \quad I$$

$$2 \times \overrightarrow{CZ} + 2 \times \overrightarrow{AX} + 2 \times \overrightarrow{BY} = 0$$

$$2 \times (\overrightarrow{CZ} + \overrightarrow{AX} + \overrightarrow{BY}) = 0$$

$$(\overrightarrow{CZ} + \overrightarrow{AX} + \overrightarrow{BY}) = 0 \quad II$$

$$\overrightarrow{AY} = \overrightarrow{CY} - \overrightarrow{CA}$$

$$\overrightarrow{BZ} = \overrightarrow{AZ} - \overrightarrow{AB}$$

$$\overrightarrow{CX} = \overrightarrow{BX} - \overrightarrow{BC}$$

By adding

$$\begin{aligned} \overrightarrow{AY} + \overrightarrow{BZ} + \overrightarrow{CX} &= (\overrightarrow{CY} - \overrightarrow{CA}) + (\overrightarrow{AZ} - \overrightarrow{AB}) + (\overrightarrow{BX} - \overrightarrow{BC}) \\ &= (\overrightarrow{CY} + \overrightarrow{AZ} + \overrightarrow{BX}) - (\overrightarrow{CA} + \overrightarrow{AB} + \overrightarrow{BC}) \\ &= 0 + 0 \\ &= 0 \end{aligned}$$



Further Work with Vectors Solution

$$(i) \quad \vec{r} = k\overrightarrow{AB} + \overrightarrow{OA}$$

$$\overrightarrow{OA} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

$$\overrightarrow{AB} = \begin{bmatrix} 4-2 \\ -3-(-1) \\ 2-4 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\vec{r} = k(2) \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

$$\begin{aligned} (ii) \quad \text{Line 1} \quad x &= 2 + \lambda & \text{--- I} \\ y &= -1 - \lambda & \text{--- II} \\ z &= 4 - \lambda & \text{--- III} \end{aligned}$$

$$\begin{aligned} \text{Line 2} \quad \frac{x+3}{k} &= \frac{4-y}{6} & \text{--- A} \\ \frac{4-y}{6} &= z-1 & \text{--- B} \end{aligned}$$

Substitute Equations II and III into equation B:

$$\begin{aligned} \frac{4 - (-1 - \lambda)}{6} &= 4 - \lambda - 1 \\ 5 + \lambda &= 18 - 6\lambda \quad \lambda = \frac{13}{7} \end{aligned}$$

Substitute equations I and II into equation A:

$$\begin{aligned} \frac{2 + \lambda + 3}{k} &= 4 - \lambda - 1 \\ 5 + \lambda &= k(3 - \lambda) \end{aligned}$$

$$k = \frac{5 + \lambda}{3 - \lambda}$$

Substituting the value for λ

$$k = \frac{5 + \frac{13}{7}}{3 - \frac{13}{7}} = 6$$



Further Work with Vectors Solution

$$\begin{aligned} (i) \quad \overrightarrow{OP} &= \overrightarrow{OA} + \overrightarrow{AP} \\ &= 2\hat{a} + \lambda \overrightarrow{AB} \\ &= 2\hat{a} + \lambda (\overrightarrow{OB} - \overrightarrow{OA}) \\ &= 2\hat{a} + \lambda (2\hat{b} - 2\hat{a}) \\ &= 2\hat{a} + 2\lambda\hat{b} - 2\lambda\hat{a} \\ &= 2(1-\lambda)\hat{a} + 2\lambda\hat{b} \end{aligned}$$

$$\begin{aligned} (ii) \quad \overrightarrow{OP} &= \overrightarrow{OM} + \overrightarrow{MP} \\ &= \frac{1}{2}\overrightarrow{OB} + \mu \overrightarrow{MN} \\ &= \frac{1}{2}\hat{b} + \mu (\overrightarrow{ON} - \overrightarrow{OM}) \\ &= \frac{1}{2}\hat{b} + \mu (5\hat{OA} - \frac{1}{2}\hat{OB}) \\ &= \frac{1}{2}\hat{b} + 10\mu\hat{a} - \mu\hat{b} \\ &= 10\mu\hat{a} + (1-\mu)\hat{b} \end{aligned}$$

$$\begin{aligned} (iii) \quad 2(1-\lambda)\hat{a} + 2\lambda\hat{b} &= 10\mu\hat{a} + (1-\mu)\hat{b} \\ \text{As } \hat{a}, \hat{b} \text{ independent (not parallel),} \\ 2(1-\lambda) &= 10\mu, \quad 2\lambda = 1-\mu \\ \lambda + 5\mu &= 1 \quad \text{--- (1)} \quad 2\lambda + \mu = 1 \quad \text{--- (2)} \\ 2 \times \text{(1)} - \text{(2)}: \quad 9\mu &= 1 \Rightarrow \mu = \frac{1}{9} \\ \text{Sub in (1)}: \quad 2\lambda + \frac{1}{9} &= 1 \Rightarrow \lambda = \frac{4}{9} \end{aligned}$$

$$\text{ie. } \overrightarrow{AP} = \frac{4}{9} \overrightarrow{AB}$$

$$\text{ie. } P \text{ divides } \overrightarrow{AB} \text{ in ratio } \boxed{4:5}$$



Further Work with Vectors Solution

$$\begin{aligned} (i) \quad 9 - 8\lambda &= -16 + \mu \\ 8\lambda + \mu &= 25 \\ 4 + 5\lambda &= 10 + 9\mu \\ 5\lambda - 9\mu &= 6 \\ \text{Solving simultaneously: } \lambda &= 3, \mu = 1 \\ 2 - 3\lambda &= \alpha + 4\mu \\ -7 &= \alpha + 4 \\ \alpha &= -11 \end{aligned}$$

$$\overrightarrow{OR} = \begin{bmatrix} 9 \\ 2 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} -8 \\ -3 \\ 5 \end{bmatrix} = \begin{bmatrix} -15 \\ -7 \\ 19 \end{bmatrix}$$

$$(ii) \quad \text{if } \vec{a} = \begin{bmatrix} -8 \\ 3 \\ 5 \end{bmatrix}, \vec{b} = \begin{bmatrix} -4 \\ 1 \\ 9 \end{bmatrix} \quad \text{then } \vec{a} \cdot \vec{b} = -8 + 12 + 45 = 49$$

$$|\vec{a}| = 7\sqrt{2}, \quad |\vec{b}| = \sqrt{85}$$

$$\cos \theta = \frac{49}{7\sqrt{2} \cdot \sqrt{85}} = \frac{1}{2}$$

$$\theta = 60^\circ$$

$$(iii) \quad \text{Diagram showing vector BM and angle } 60^\circ. \quad BM = 10 \sin 60^\circ = 5\sqrt{3} \text{ units}$$

